Alphabet Soup: CLT, SE, and CI

Code

STAT 4714

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0.1 Quick Recap

- **Repeated sampling** can be used to make inference.
- **Sampling distributions** describe how a statistic varies over hypothetical repetitions.
- Sampling distributions get **narrower** as n increases (smaller SE), as we saw in various simulations.

0.2 Goals for Today

- Build intuition for the **Central Limit Theorem (CLT)**.
- Understand **why CLT matters** for inference.
- Define and use **Standard Error (SE)** for common estimators.
- Connect SE to **margin of error** and **confidence intervals**.

0.3 Why Learn the CLT?

- **Goal:** Make inference using sampling distributions.
- **Problem:** Exact sampling distributions are often **messy or unknown** (skewed data, odd estimators, etc.).
- **Solution (CLT):** For many settings with finite variance & independence, **averages ≈ Normal** as n grows.
  - With the normal distribution as the reference distribution, we unlock a lot of things for free: convenient formulas for standard error, confidence intervals, and various tests.

1.1 Standardized Sample Means

We might **standardize** the sample mean $\bar{X}$ to compare it to a *universal* reference distribution (i.e., Normal or t-distribution), which lets us make scale-free inference.

- The **standardized sample mean** is

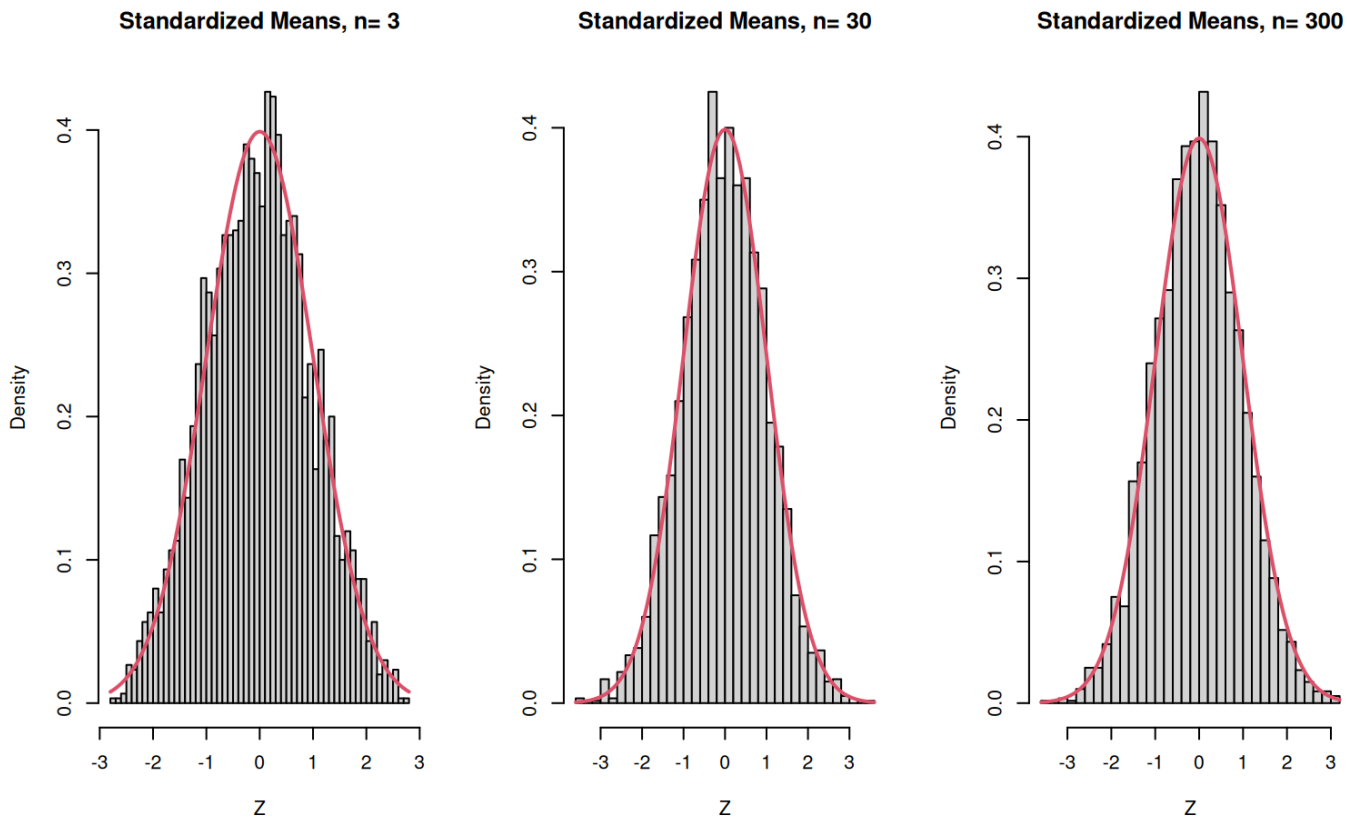
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

1.2 CLT: The Idea

For many populations with finite variance, the distribution of the **standardized sample mean** becomes approximately **Normal(0,1)** as sample size n grows.

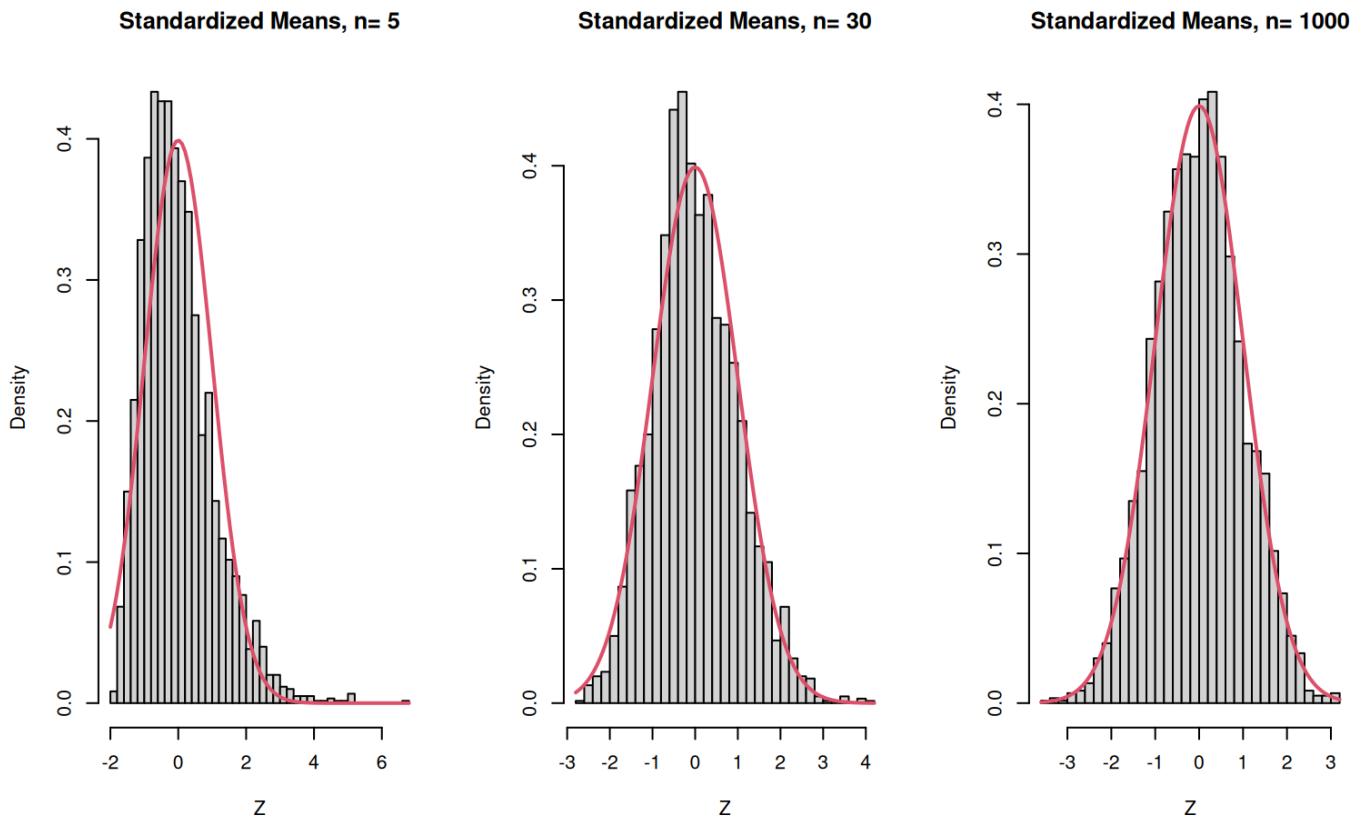
- Requires independence (or weak dependence) and finite variance.
- We can use **Normal-based inference** without knowing the population distribution.
- Works even if the **data** are not normal.
  - Let's see some examples…

1.3 Simulation: Uniform(0,1)



Histograms of standardized means from Uniform(0,1), which approach N(0,1) as n increases.

1.4 Simulation: Exponential(1)

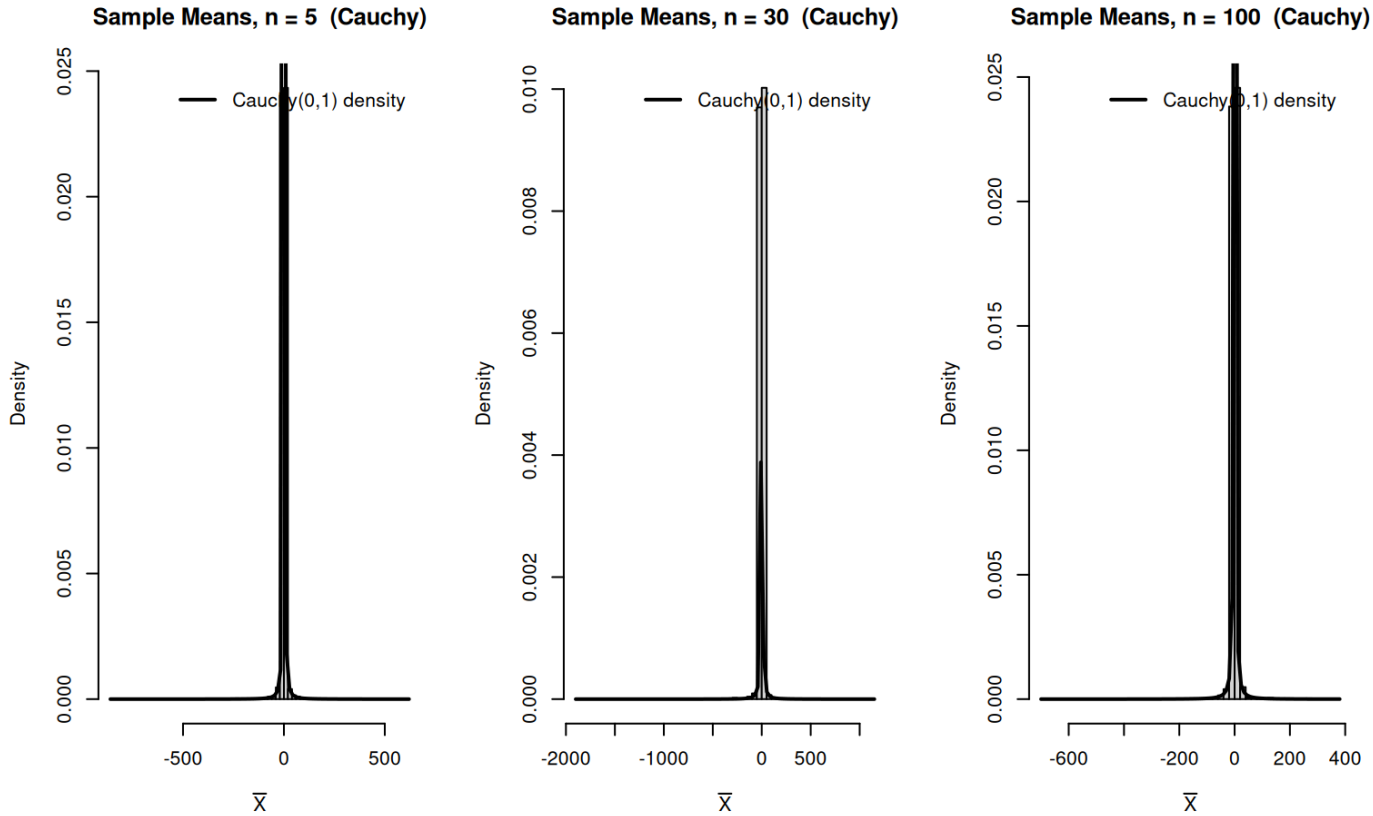


Histograms of standardized means from Exponential(1). Slower convergence.

1.5 When CLT Fails

If variance is **infinite** (e.g., Cauchy), the CLT does **not** apply.

- The mean can be extremely unstable.
- No amount of sample size guarantees normality.



Histograms of sample means from Cauchy(0,1). No convergence to Normal.

1.6 CLT Takeaways

- Great for **sample means** when variance is finite and independence is reasonable.
- CLT logic assumes **independent** observations from the same process; dependence (time/space/clusters) can break the approach.
- **Heavy tails** result in slow convergence because outliers can dominate.
- Approximating with a standard normal distribution unlocks many convenient formulas. Let's see one now!

2.1 What is SE?

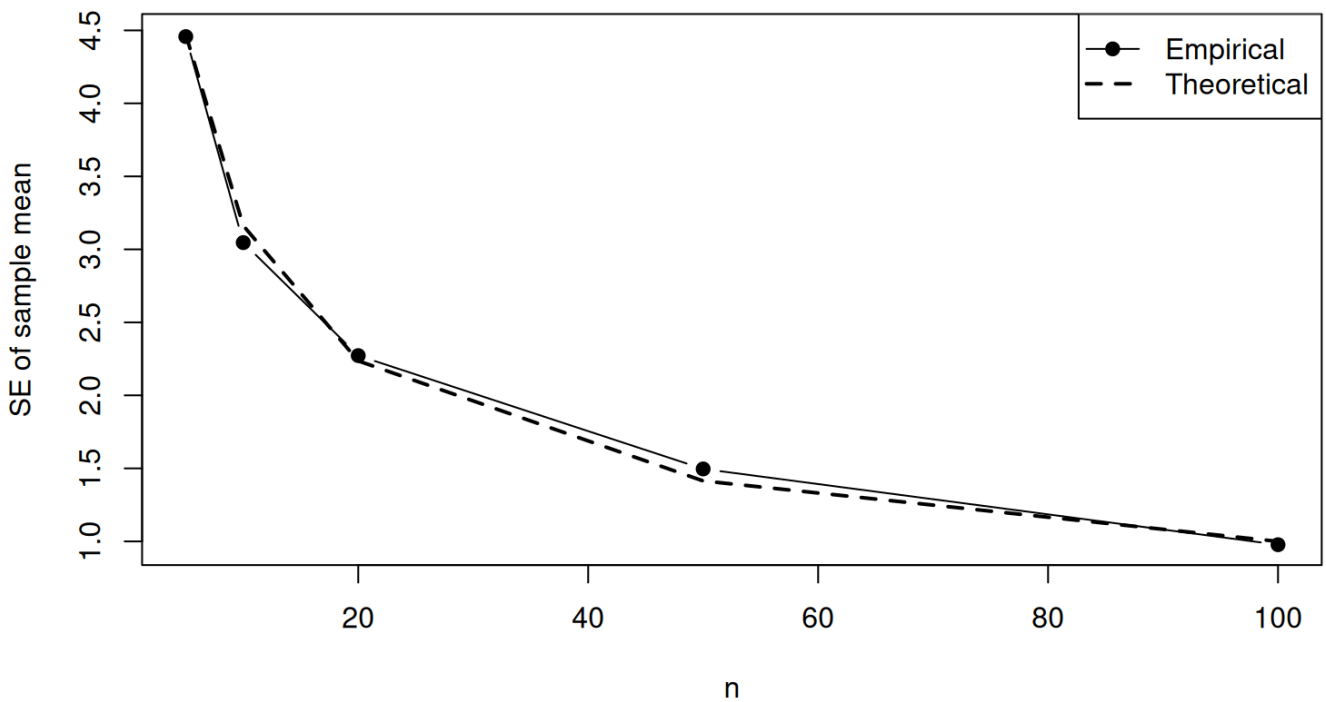
Standard Error (SE) is the standard deviation of a statistic's sampling distribution. For some cases, we know the exact formula:

- For a sample mean: $SE(\bar{X}) = \sigma/\sqrt{n}$.
  - Note that this is the denominator of the CLT transformation!
- For a sample proportion: $SE(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$.

2.2 What is SE?

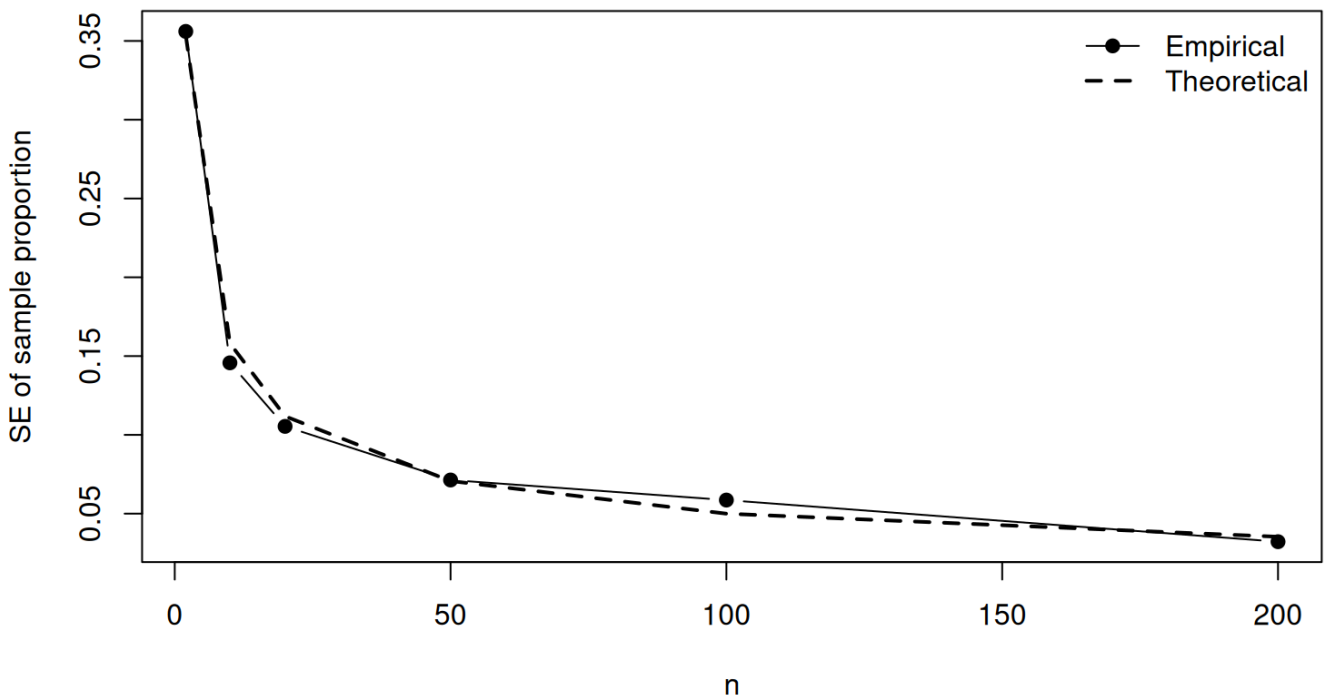
- In many cases, we may not know the exact formula (or may not want to work with it).
- For any statistic, we can estimate variability/SE via **resampling**.
  - A popular approach is *bootstrapping*, which we will cover in a future lecture.
- We can compare the true (theoretical) SE with its empirical estimate, where the estimate is obtained by `sqrt(var(sample_means))` …

2.3 Simulation: SE of the Mean



Empirical vs theoretical SE for Normal(50,10).

2.4 Simulation: SE of the Proportion



Empirical vs theoretical SE for Binomial(n=1,p=0.5).

2.5 Difference of Means

Motivation: Engineers often compare two designs/components and might be interested in the statistic: $\bar{X}_1 - \bar{X}_2$, which is a **difference of means**.

- The SE of $\bar{X}_1 - \bar{X}_2$ tells us how much *sampling variation* to expect in the difference.
- For independent designs with unequal, unknown variances, the (Welch) SE is

$$\sqrt{s_1^2/n_1 + s_2^2/n_2},$$

  where s_1^2 and s_2^2 are the sample variances of the designs.

2.6 Wait, There's More!

- Assuming independent samples, we typically use a Welch t test. If measurements are paired, we analyze the differences with $SE(\bar{D}) = s_D/\sqrt{n}$, where

$$s_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}(D_i - \bar{D})^2, \quad D_i = X_{1i} - X_{2i}.$$

- If $\sigma_1^2 = \sigma_2^2$ is justified, we can use a pooled SE: $SE(\bar{X}_1 - \bar{X}_2) = s_p\sqrt{1/n_1 + 1/n_2}$, where

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

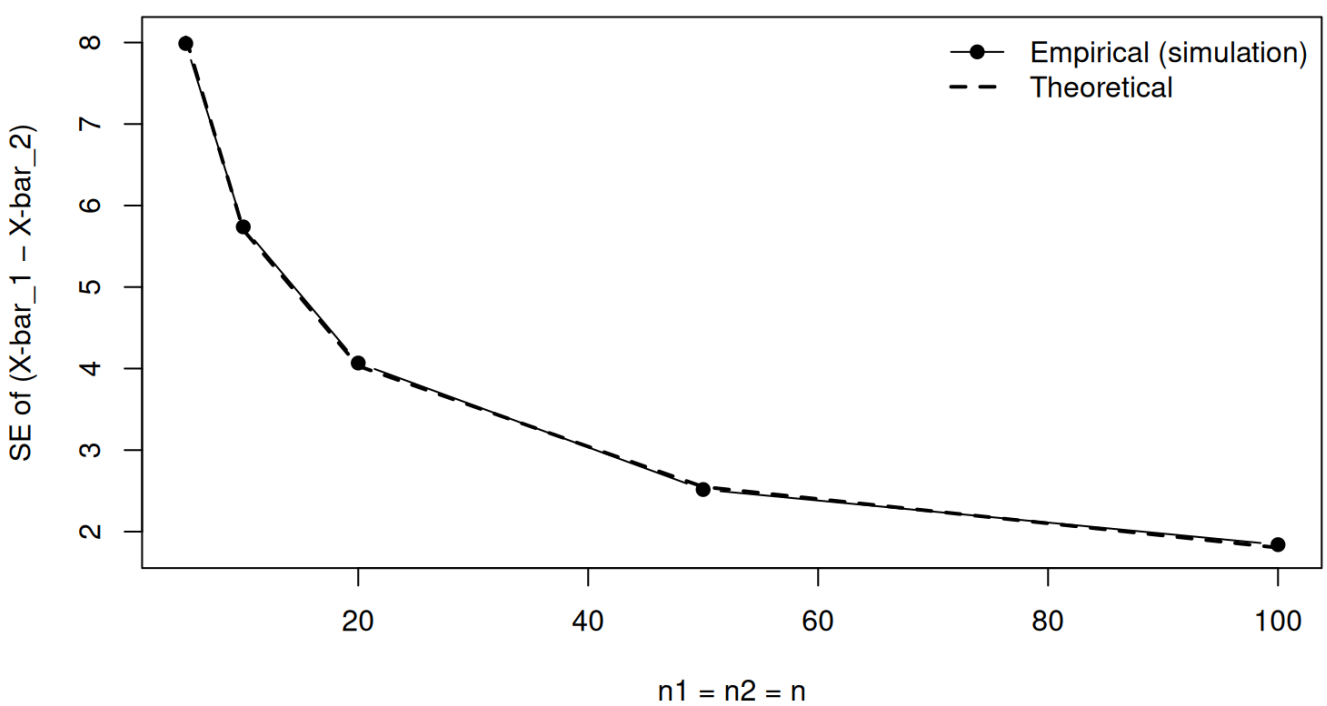
-

2.7 Difference of Means

How can such a simple extension be so complicated? Do we need to memorize all these formulas?

- Not really – we can always simulate to approximate the sampling distribution. We just need to ensure that our simulation mirrors the study design!
- If we adequately approximate the sampling distribution, then we can adequately estimate SE.
- Constructing the sampling distribution for $\bar{X}_1 - \bar{X}_2$ is as simple as repeatedly sampling both designs, computing their sample means and subtracting. It's no more complicated than what we've been doing…

2.8 Simulation: SE of Difference of Means



Empirical vs theoretical SE of the mean difference with unequal variances.

2.9 Formulas vs Simulation

- **Simulation wins** when formulas are messy, assumptions are dubious, or designs are complex (unequal variances, non-Normal, small samples, skew/heavy tails, odd estimators).
- **Formulas win** for **speed, planning, and transparency** (power/size calculations, analytic checks, closed-form insight).
- **Key idea:** simulate the **study mechanism** (how data are generated/collected), then compute your statistic each time. The histogram is your sampling distribution, which easily unlocks concepts like SE.

2.10 Formulas vs Simulation

Both **simulation** and **formulas** require the **correct design/model**.

- **Simulation:** *Pros:* flexible, few formulas; *Cons:* needs enough replication, careful resampling scheme.
- **Analytic:** *Pros:* fast, transparent; *Cons:* wrong if model/design is misspecified.

3.1 Motivating CIs

- A point estimate (e.g., sample mean) is just one number; however, every sample gives a slightly different estimate.
- Just like we need mean and variance to adequately describe a population distribution, we should also report uncertainty in our estimate.
- The SE tells us the average variation of the estimate. The **confidence interval (CI)** translates that uncertainty into a range of plausible values for the true parameter, tied to a chosen confidence level. They help us understand where the true value might be and how uncertain we are.

3.2 Putting the Confidence in CI

- The width of CIs are determined by our desired level of confidence. We typically report $(1-\alpha)$ confidence intervals, where α is the **significance level** we want (e.g., 95% confidence with $\alpha = 1 - 0.95 = 0.05$). Thus, the CI is a function of α.
- Higher confidence leads to wider intervals because we need more coverage.

3.3 CIs from Sampling Distributions

We can obtain the $(1-\alpha)$ CI from a sampling distribution using quantiles:

$$\widehat{\mathrm{CI}} = [\, q_{\alpha/2},\ q_{1-\alpha/2} \,],$$

where q_c is the cth quantile of the sampling distribution.

▼ Code

```
set.seed(123)
pop <- rnorm(100000, mean = 12, sd = 5)
n <- 30
B <- 5000
sample_means <- replicate(B, mean(sample(pop, n)))
ci <- quantile(sample_means, c(0.025, 0.975)) # alpha=0.05
ci
```

```
    2.5%    97.5%
10.24639 13.81338
```

3.4 Interpreting CIs

- CIs are notoriously tricky to interpret. A 95% CI of $[10, 14]$ means that 95% of the CIs we'd construct would contain the true parameter. It does **not** mean that there is a 95% probability that the true value is in $[10, 14]$!
  - *Fun fact*: This is a common example of how *Bayesian* statistics is more intuitive than frequentist statistics (which is what we're learning)!
- Importantly, the truth is fixed, and the intervals change from sample to sample.

3.5 A Related Concept: MOE

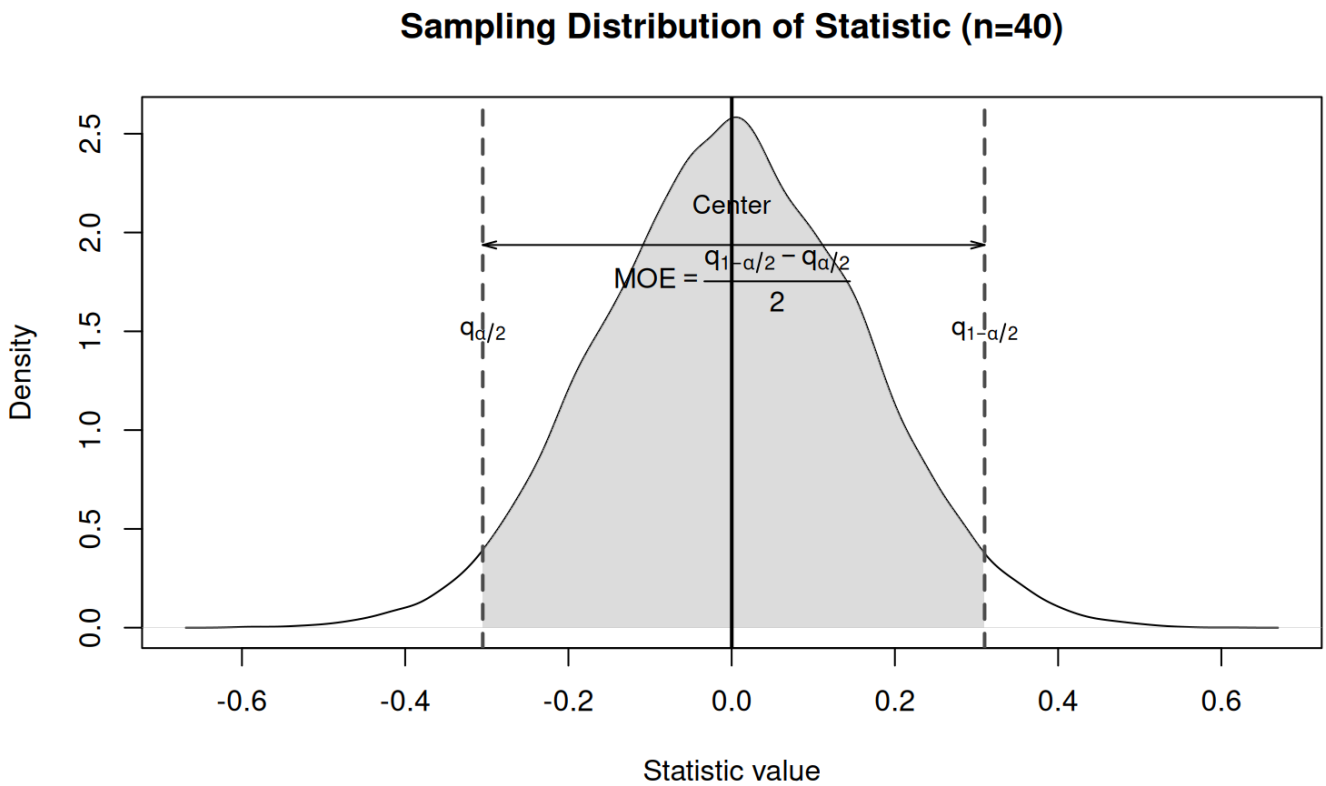
- The **confidence interval** for a statistic T with a symmetric sampling distribution is

$$CI = [T - MOE, T + MOE],$$

  where the **margin of error (MOE)** is a scaled version of SE.
- If we simulate the **sampling distribution**, then

$$\widehat{\text{CI}} = [\, q_{\alpha/2},\, q_{1-\alpha/2}\,], \quad \underbrace{\widehat{\text{MOE}} = \tfrac{1}{2}\big(q_{1-\alpha/2} - q_{\alpha/2}\big)}_{\text{requires symmetry}}.$$

3.6 Estimating MOE with Simulation



MOE is the half-width between central quantiles of the sampling distribution.

3.7 Asymmetric Sampling Distribution?

- Then, we cannot use the confidence interval approach to compute MOE.
- Honestly, most people just report confidence intervals, which can always be obtained as

$$\widehat{\text{CI}} = [\, q_{\alpha/2},\ q_{1-\alpha/2}\,].$$

- Under symmetric sampling distributions, MOE is a convenient metric that reports the CI as a single quantity… but it's outdated.

3.8 Recipe: Sampling → CI → MOE

1. Simulate the statistic T (e.g., sample mean) many times under the correct model.
2. Compute the quantiles of the sampling distribution.
3. Compute $MOE = \frac{1}{2}(q_{1-\alpha/2} - q_{\alpha/2})$.

▼ Code

```
set.seed(1)
pop <- rnorm(100000, mean = 50, sd = 10)
```

```r
n <- 30; B <- 5000; alpha <- 0.05
sample_means <- replicate(B, mean(sample(pop, n)))
SE_hat <- sd(sample_means)
qs <- quantile(sample_means, c(alpha/2, 1 - alpha/2))
MOE_hat <- 0.5 * diff(qs)
list(SE = SE_hat, CI = as.numeric(qs), MOE = MOE_hat)
```

```
$SE
[1] 1.84161

$CI
[1] 46.41452 53.60933

$MOE
   97.5%
3.597405
```

4.1 Pitfalls

- If there is hidden **dependence** in the data, then SE is underestimated.
- **Heavy tails/outliers** result in slow CLT convergence.
- For skewed data, consider **sample median** over sample mean.

4.2 Summary

- CLT explains why normal-based inference works for many “average-like” statistics.
- SE is the variability of an estimator; can be estimated via simulation.
- CIs are an efficient way to summarize uncertainty associated with estimates at specified confidence levels.
    - For symmetric sampling distributions, we saw the connection between SE/MOE and CIs.

4.3 Breakout Work

1. You measure latency on two groups of 20 devices each. The difference in means are right-skewed with a few outliers. How might you estimate SE via simulation?
2. If we want narrower confidence intervals, what can we change about our **study design**?
3. Why are **confidence intervals** important for engineers designing experiments or quality checks?
4. When would you **avoid CLT**?